Week 8

nano minishell.c

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <errno.h>

#define MAX_LINE 1024
#define MAX_HISTORY 100
#define MAX_CMDS 10
#define MAX_ARGS 64

char history[MAX_HISTORY][MAX_LINE];
int history_count = 0;

void save_history(const char *line) {
    int pos = history_count % MAX_HISTORY;
    snprintf(history[pos], MAX_LINE, "%s", line);
    history_count++;
}
```

```
void show_history(void) {
    int start = history_count > MAX_HISTORY
                ? history_count - MAX_HISTORY : 0;

    for (int i = start; i < history_count; i++) {
        printf("%d  %s\n", i + 1,
               history[i % MAX_HISTORY]);
    }
}

int parse_command(char *s, char **args) {
    int n = 0;
    char *token = strtok(s, " \t\n");

    while (token && n < MAX_ARGS - 1) {
        args[n++] = token;
        token = strtok(NULL, " \t\n");
    }

    args[n] = NULL;
    return n;
}

void execute_pipeline(char *line) {
    char *commands[MAX_CMDS];
    char *args[MAX_CMDS][MAX_ARGS];
    int pipes[MAX_CMDS - 1][2];
    int count = 0;
```

```
char *saveptr;
char *part = strtok_r(line, "|", &saveptr);

while (part && count < MAX_CMDS) {
    commands[count++] = part;
    part = strtok_r(NULL, "|", &saveptr);
}

if (part || count == 0) {
    fprintf(stderr, "Error: invalid or too many commands\n");
    return;
}

for (int i = 0; i < count; i++) {
    if (parse_command(commands[i], args[i]) == 0) {
        fprintf(stderr, "Error: empty command in pipeline\n");
        return;
    }
}

for (int i = 0; i < count - 1; i++) {
    if (pipe(pipes[i]) == -1) {
        perror("pipe");
        for (int j = 0; j < i; j++) {
            close(pipes[j][0]);
            close(pipes[j][1]);
        }
```

```
        return;
    }
}

pid_t pids[MAX_CMDS];
int launched = 0;

for (int i = 0; i < count; i++) {
    pid_t pid = fork();

    if (pid == -1) {
        perror("fork");
        break;
    }

    if (pid == 0) {
        if (i > 0 &&
            dup2(pipes[i - 1][0], STDIN_FILENO) == -1) {
            perror("dup2 input");
            _exit(1);
        }

        if (i < count - 1 &&
            dup2(pipes[i][1], STDOUT_FILENO) == -1) {
            perror("dup2 output");
            _exit(1);
        }
```

```
            for (int j = 0; j < count - 1; j++) {
                close(pipes[j][0]);
                close(pipes[j][1]);
            }

            execvp(args[i][0], args[i]);
            perror(args[i][0]);
            _exit(127);
        }

        pids[launched++] = pid;
    }

    for (int i = 0; i < count - 1; i++) {
        close(pipes[i][0]);
        close(pipes[i][1]);
    }

    for (int i = 0; i < launched; i++) {
        int status;
        while (waitpid(pids[i], &status, 0) == -1) {
            if (errno != EINTR) {
                perror("waitpid");
                break;
            }
        }
    }
}
```

```
int main(void) {
    char line[MAX_LINE];

    printf("==================================\n");
    printf(" WEEK-8 CUSTOM LINUX SHELL\n");
    printf(" Type 'history' or 'exit'\n");
    printf("==================================\n");

    while (1) {
        printf("week8$ ");
        fflush(stdout);

        if (!fgets(line, sizeof(line), stdin)) {
            break;
        }

        line[strcspn(line, "\n")] = '\0';

        if (line[0] == '\0') {
            continue;
        }

        save_history(line);

        if (strcmp(line, "exit") == 0) {
            break;
        }
```

```
    if (strcmp(line, "history") == 0) {
        show_history();
        continue;
    }

    execute_pipeline(line);
  }

  printf("Shell closed.\n");
  return 0;
}
```

```
bharath@DESKTOP-6D34C2R:~/week8-ossp$ nano minishell.c
bharath@DESKTOP-6D34C2R:~/week8-ossp$ gcc -Wall -Wextra -std=c11 minishell.c
 -o minishell
bharath@DESKTOP-6D34C2R:~/week8-ossp$ ./minishell
================================
 WEEK-8 CUSTOM LINUX SHELL
 Type 'history' or 'exit'
================================
week8$ ls | wc -l
2
week8$ cat /etc/passwd | grep root | wc -l
1
week8$ history
1  ls | wc -l
2  cat /etc/passwd | grep root | wc -l
3  history
week8$ ls
pwd
whoami
historyminishell  minishell.c
week8$ /home/bharath/week8-ossp
week8$ bharath
week8$
1  ls | wc -l
2  cat /etc/passwd | grep root | wc -l
3  history
4  ls
5  pwd
6  whoami
7  history
week8$
```